

Weekly Report

PPO (Proximal Policy Optimization)

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Contents

1	Background	1
2	Model Formation	1
3	Loss Functions	1
3.1	Clipped Surrogate Loss	1
3.2	Value Function Loss	2
3.3	Entropy Bonus	2
3.4	Final loss function	3

1 Background

In reinforcement learning that use neural networks (NN), the policy function, which maps states to actions or action probabilities or so on, is often represented by a neural network. Traditionally, the new name **PPO** means modified loss function during the training process, and that is same. The loss function of PPO is consisting of three parts [1].

1. **Clipped Surrogate Loss** (L_t^{CLIP})

Activating good actions and deactivating bad actions, but not too much change in unit step.

2. **Value Function Loss** (L_t^{VF})

Error between the predicted state value function and the actual state value.

3. **Entropy Bonus** ($S[\pi_\theta](s_t)$)

Encouraging exploration by adding entropy to the policy.

2 Model Formation

Actually, the detailed structure of model is not important. The formation of model is two parts.

1. **Actor** (π_θ)

This is the policy function that maps states (s) to actions (a) as probabilities. If the action space is discrete, the output of actor is logits of each action.

2. **Critic** (V_ϕ)

This supplementary part is the value function that maps states (s) to state values ($V(s)$).

Ordinarily, the actor and critic share the same NN structure.

3 Loss Functions

3.1 Clipped Surrogate Loss

Let's define the advantage function A_t in Equation 1. For positive case, which means the actions is good, NN should recommend that action. However, for negative case, which means the actions is bad, NN should not recommend that action.

$$\begin{aligned} A_t &:= R_t - V(s_t) \in \mathbb{R} \\ A_t > 0 &\Rightarrow \pi_\theta(a_t|s_t) \uparrow \\ A_t < 0 &\Rightarrow \pi_\theta(a_t|s_t) \downarrow \end{aligned} \tag{1}$$

In addition, the probability ratio $r_t(\theta)$ is defined in Equation 2. For larger than 1, the new model is more likely to recommend that action than before. In other case, if the ratio is smaller than 1, the new model is less likely to recommend that action than before. The initial $r_t(\theta)$ is 1, which means the new model is same as before.

$$r_t(\theta) := \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \in \mathbb{R} \tag{2}$$

Let's consider the product of $r_t(\theta)$ and A_t in Equation 3. $r_t(\theta)$ roles as a gradient direction, and A_t roles as magnitude. To maximize the reward, the product should be maximized.

$$r_t(\theta)A_t \in \mathbb{R} \quad (3)$$

1. **Case 1:** $A_t > 0$

That action is good, so $r_t(\theta)$ should be larger than 1. Then, the product become larger.

2. **Case 2:** $A_t < 0$

That action is bad, so $r_t(\theta)$ should be smaller than 1. Then, the product become moving the maximizing direction of that product.

However, to prevent the large change of $r_t(\theta)$, the clipped method is used.

$$\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t) \in \mathbb{R} \quad (4)$$

For N batch data (piece of episode; based on the Markov premises), the clipped surrogate loss is defined in Equation 5.

$$\begin{aligned} L^{CLIP}(\theta) &= \frac{1}{N} \sum_{i=1}^N \min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t) \\ &= \hat{\mathbb{E}}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)] \end{aligned} \quad (5)$$

3.2 Value Function Loss

The role of second loss is to improve critic (V_ϕ). In the advantage function in Equation 1, the difference between the actual state value and the predicted state value is used in Equation 6. Therefore, the critic should be smart to precisely predict the input state value.

$$\begin{aligned} V(s_t) &\approx r_t + \gamma V_\phi(s_{t+1}) \\ &= \sum_{a_t \in \mathcal{A}} \pi(a_t|s_t) \sum_{s_{t+1} \in \mathcal{S}} P(s_{t+1}|s_t, a_t) [r(s_t, a_t) + \gamma V_\phi(s_{t+1})] \end{aligned} \quad (6)$$

So, the value function loss is defined in Equation 7. The mean squared error (MSE) is used to minimize the difference between the actual state value and the predicted state value. The objective is to minimize $L^{VF}(\phi)$.

$$\begin{aligned} L^{VF}(\phi) &= \frac{1}{N} \sum_{t=1}^N (V_\phi(s_t) - (r_t + \gamma V_\phi(s_{t+1})))^2 \\ &= \hat{\mathbb{E}}_t [(V_\phi(s_t) - (r_t + \gamma V_\phi(s_{t+1})))^2] \end{aligned} \quad (7)$$

3.3 Entropy Bonus

The role of third loss is to encourage variable exploration, not to be stuck in local optimums. The high entropy means the probability distribution is approximately uniform. To encourage exploration, the entropy should be kept high. The entropy bonus is defined in Equation 8. The objective is to maximize it.

$$\begin{aligned}
S[\pi_\theta](s_t) &= - \sum_{a_t \in \mathcal{A}} \pi_\theta(a_t|s_t) \log \pi_\theta(a_t|s_t) \\
L^S(\theta) &= \frac{1}{N} \sum_{t=1}^N S[\pi_\theta](s_t) \\
&= \hat{\mathbb{E}}_t [S[\pi_\theta](s_t)]
\end{aligned} \tag{8}$$

3.4 Final loss function

The final loss function is defined in Equation 9 with linear combination. The objective is to minimize $L(\theta, \phi)$ with appropriate scalar parameters c_1 and c_2 .

$$L(\theta, \phi) = -L^{CLIP}(\theta) + c_1 L^{VF}(\phi) - c_2 L^S(\theta) \tag{9}$$

References

- [1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017).